

0.1 RQ7-Specific Coding Codebook

Coding purpose. RQ7 examines which challenges the included reviews emphasize, which future directions they advocate, how they frame the field’s priorities overall, and whether they register a shift in those priorities.

Structure of RQ7 coding. RQ7 differs from the other research questions. It combines three kinds of record: ordinal item scores, categorical framing codes, and binary shift signals. Only the first uses the shared 0–3 scale.

Challenge and future-direction items are scored 0–3 on the nine challenge codes C1 to C9 and the eleven future-direction codes F1 to F11, defined in Table 1.

Framing codes are categorical. They record the review’s dominant and secondary challenge family, its emphasis on safety and trust, its dominant future orientation, and the overall strength of its RQ7 contribution.

Shift signals are binary. They record whether the review registers a change in research priorities along four axes.

Item codes. The nine challenge codes and eleven future-direction codes are listed below. The same definitions appear in the main text; they are repeated here so that this codebook can be applied without reference to it.

Table 1: Challenge and future-direction item codes.

Code	Block	Description
C1	Challenge	Simulation dependence or weak real-building evidence
C2	Challenge	Training burden or costly trial-and-error learning
C3	Challenge	State or action formulation complexity
C4	Challenge	Multi-objective complexity
C5	Challenge	Generalization or transfer problems
C6	Challenge	Scalability and multi-agent coordination
C7	Challenge	Operational and implementation barriers
C8	Challenge	Safety or constraint risk
C9	Challenge	Trust, privacy, governance, or ethics concerns
F1	Future direction	Deep RL or advanced algorithms
F2	Future direction	Offline or model-based RL
F3	Future direction	Transfer learning across buildings or tasks
F4	Future direction	Hierarchical or multi-timescale RL
F5	Future direction	Meta-RL or online adaptation
F6	Future direction	Multi-agent RL
F7	Future direction	Hybrid MPC and RL
F8	Future direction	Real-world deployment pathways
F9	Future direction	Occupant-aware or human-centric control
F10	Future direction	Benchmarking or open infrastructure
F11	Future direction	Explainability, privacy, governance, or ethics

Boundary rules between adjacent codes. Four pairs of codes are close enough to need an explicit rule.

F3 covers transfer of policies or models between buildings, zones, or tasks. F5 covers meta-learning and online adaptation of a controller to changing conditions. A review that advocates transfer without discussing adaptation scores on F3 only. Where a review treats both, both are scored.

C5 covers generalization and transfer as a stated problem. F3 covers transfer as an advocated solution. A review can score highly on C5 while scoring 0 on F3 if it names the difficulty without proposing transfer as a remedy.

C1 covers weak real-building evidence as a stated problem. F8 covers pathways to real-world deployment as an advocated direction. The same separation between problem and remedy applies.

C9 and F11 cover the same subject matter, trust, privacy, governance, and ethics, as a challenge and as a future direction respectively.

What the scale measures. The 0–3 scale records treatment intensity, meaning how prominently a review emphasizes a challenge or advocates a future direction within its own account of the literature. It does not measure how severe a challenge is, how promising a direction is, or how much primary work exists on either. A score of 3 on a challenge code means the review foregrounds that challenge, not that the challenge is unresolved. The score describes the review, not the field.

Table 2: Treatment-intensity scale as applied to the C and F item codes.

Score	Label	General interpretation
0	Not addressed	The challenge or direction is not identifiable in the review.
1	Mentioned	It is named or listed without development, for example as one item in a closing list of open problems.
2	Discussed	It is explained, with its significance for building control made clear, but it does not organize part of the synthesis.
3	Structuring role	It organizes part of the review’s synthesis, through a dedicated section, a recurring theme, or an explicit basis for the review’s agenda.

Framing codes. Framing codes are assigned once per review from a controlled set of values. They are not ordinal and are not aggregated with the item scores.

Table 3: Categorical framing codes and their value sets.

Framing code	Values and assignment rule
Dominant challenge family	Technical-dominant; operational or implementation-dominant; technical-operational hybrid; technical and human-centric hybrid. Assigned to the family carrying the review’s highest and most sustained challenge emphasis. A hybrid value is used when two families carry comparable weight.
Secondary challenge family	Operational or implementation; safety, trust, and governance; technical and generalization. Assigned to the family carrying the next strongest emphasis after the dominant one.
Safety and trust emphasis	Limited; moderate; strong. Intermediate values are recorded where the review sits between two levels. Assigned from the combined treatment of safety, constraint risk, trust, privacy, and governance.
Dominant future orientation	Free text. A short phrase naming the direction the review’s agenda points toward. This is a narrative summary and not a coded value.
Overall RQ7 contribution strength	Strong; moderate. Records how substantively the review supports RQ7 as a whole.

Shift signals. Four binary signals record whether a review registers a change in research priorities. A signal is coded 1 when the review states or clearly implies that the field’s emphasis is moving along that axis, and 0 when it does not. Noting that both poles exist is not sufficient; the review must indicate movement.

Table 4: Binary shift signals.

Signal	Coded 1 when the review indicates a move
Simulation to real-world deployment	From simulation-based evaluation toward deployment in real buildings.
Single building to community or larger-scale coordination	From single-building control toward multi-building, community, or larger-scale coordination. Reviews focused on individual dwellings are coded on the same axis.
Pure optimization to human-centered framing	From purely technical objectives toward occupant-centered or stakeholder-centered framing.
Performance to safety, trust, and governance	From performance maximization toward safety, trust, privacy, or governance concerns.

Label normalization. The second shift signal is recorded in the audit tables under two labels, single-home and single-building. Both refer to the same axis and are reported here as one signal. Normalization changes labels only and leaves all values unchanged.

Zero and applicability. A score of 0 on an item code means the review does not address that challenge or direction. The scale does not separate a topic that a review considered and did not develop from one that fell outside its declared scope. Both are recorded as 0. A score of 0 should therefore not be read as evidence that a review overlooked something it could have covered.

A shift signal of 0 means the review does not register movement along that axis. It does not mean the review denies that movement is occurring.

All coded reviews receive item scores, framing codes, and shift signals. RQ7 has no applicability route, so the denominator is the full coded corpus.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a challenge or direction in any of them supports a structuring score. A challenge named only in a closing list of open problems supports a score of 1. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus and coding it would make reviews non-comparable. Coding reflects the review authors' own account: a challenge reported by a cited primary study is insufficient unless the review adopts, discusses, or builds on it. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Summary fields. Each survey-level audit table also carries narrative fields recording how the review answers RQ7, its most emphasized challenge, its most emphasized future direction, and the distribution of barriers across families. These summarize the coded values in prose and are not themselves coded values.

Table 5: Worked boundary examples for RQ7 coding.

Review	Code	Value	Basis for the boundary
Mason and Gri-jalva (2019)	C8, safety or constraint risk	1	Safety appears only indirectly, through the cost and risk of physical exploration, rather than as a stated concern. Boundary against 2.
Mason and Gri-jalva (2019)	C1, simulation dependence	3	Simulation dependence and costly trial-and-error learning are foregrounded as a basis for the review’s agenda.
Mason and Gri-jalva (2019)	Dominant challenge family	Technical-dominant	The review frames the field primarily through algorithmic and formulation problems, with operational concerns secondary.
Mason and Gri-jalva (2019)	Safety and trust emphasis	Limited	Safety appears only indirectly and trust or governance is absent. Boundary against moderate.
Mason and Gri-jalva (2019)	F9, occupant-aware control	0	Human-centered control is not part of the review’s future agenda.